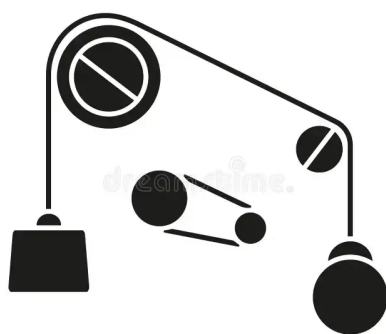




CLASSICAL MECHANICS

An Analytic Approach to Mechanics

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CHAPTER 1

Lagrange's Formalism

"No methods are necessary in this work. The methods that I lay out require neither constructions, nor geometric or mechanical reasoning, but only algebraic operations..."

Mécanique analytique
JOSEPH-LOUIS LAGRANGE

1.1 Variational Principles

Calculus of Variations

Definition 1.1 – The Variational

A variational is small, infinitesimal but *virtual* change in a quantity, and represented by δx . It is different from an infinitesimal dx in the fact the infinitesimal is a real change that occurs when an infinitesimal change in a parameter (like time t) occurs, the virtual change δx is something that occurs for no change in the parameter (i.e. $\delta t = 0$). For a functional $f(x_1, x_2, \dots x_n, t)$, the variational of the function is:

$$\delta f = \frac{\partial f}{\partial x_i} \delta x_i \quad \left(\equiv \sum_{i=1}^n \frac{\partial f}{\partial x_i} \delta x_i \right) \quad (1.1)$$

Note the use of Einstein's summation convention, which says that if an index is repeated exactly twice, then it is to be summed over all of its values. Here, we have t as the parameter upon which the functions x_i depend on, therefore it is excluded from the sum as its variational is by definition, zero. Variations commute with derivatives and integrals, i.e.

$$\delta \left(\frac{dx}{dt} \right) = \frac{d}{dt}(\delta x) \quad (1.2)$$

$$\delta \int x dt = \int \delta x dt \quad (1.3)$$

In a problem with fixed boundaries, variations vanish at these boundaries, so that the variation does not make a new function entirely, but rather is a nudge to the original function.

$$\delta x(t_1) = \delta x(t_2) = 0 \quad (1.4)$$

If the variation of functional $f(x_1, x_2, \dots, x_n, t)$ vanishes completely, then it the functional is said to be stationary, and the functions $x_i(t)$ are extremas of the functional.

Theorem 1.1 – Euler Lagrange Equation

The Euler Lagrange Equation is the solution to a specific minimization problem of functions. We define a functional $F[f(x_1, x_2, \dots, x_n, \dot{x}_1, \dot{x}_2, \dots, \dot{x}_n, t)]$ such that:

$$F[f] = \int_{t_1}^{t_2} f(x_1, x_2, \dots, x_n, \dot{x}_1, \dot{x}_2, \dots, \dot{x}_n, t) dt \quad (1.5)$$

Here, $\dot{x}_i = dx_i/dt$. The problem is to find the function f that minimizes the functional F . We do this by taking the variational δF ,

$$\delta F = \int_{t_1}^{t_2} \delta f dt = \int_{t_1}^{t_2} \frac{\partial f}{\partial x_i} \delta x_i + \frac{\partial f}{\partial \dot{x}_i} \delta \dot{x}_i dt \quad (1.6)$$

$$0 = \int_{t_1}^{t_2} \left[\frac{\partial f}{\partial x_i} - \frac{d}{dt} \left(\frac{\partial f}{\partial \dot{x}_i} \right) \right] \delta x_i dt - \frac{\partial f}{\partial \dot{x}_i} \delta x_i \Big|_{t_1}^{t_2} \quad (1.7)$$

We integrated the second term by parts to obtain the second line, and since the variational vanishes at endpoints, the term outside the integral is zero. Since we intend to find a stationary/minimum F , the variation $\delta F = 0$. We're left with the term in the integral, which has to be zero for every chosen f , and therefore, we obtain the Euler Lagrange Equation:

$$\therefore \frac{d}{dt} \left(\frac{\partial f}{\partial \dot{x}_i} \right) = \frac{\partial f}{\partial x_i} \quad (1.8)$$

The Euler Lagrange equation can be used a multitude of problems, one of which is the Brachistochrone problem.

Example 1.1 – Brachistochrone Problem

The Brachistochrone Problem was devised by Johann Bernoulli in 1696 as a challenge to the scientific community. Newton, famously, solved the problem overnight. The tools developed for the problem led to the development of the Calculus of Variations.

The problem involves finding the curve that gives us the shortest possible time to go from two points (x_1, y_1) and (x_2, y_2) upon the action of gravity. For simplicity, let those points be $O(0, 0)$ and $A(x_1, y_1)$. The problem involves finding the curve that gives us the shortest possible time to go from two points (x_1, y_1) and (x_2, y_2) upon the action

of gravity. For simplicity, let those points be $O(0,0)$ and $A(x_1, y_1)$. The time taken for an arbitrary path is:

$$t = \int_0^{t_1} \frac{1}{v} ds \quad (1.9)$$

Here, we know $v = \sqrt{2gy}$ from Newton's Laws and if we consider an infinitesimally small right triangle with hypotenuse ds and side dx and dy , then $ds = \sqrt{(dx)^2 + (dy)^2}$ which we rewrite as $ds = \sqrt{1 - (x'(y))^2} dy$ where $x'(y) = dx/dy$. So here, y is the independent parameter.

$$t = \frac{1}{\sqrt{2g}} \int_0^{t_1} \sqrt{\frac{1 - (x')^2}{2gy}} dy \quad (1.10)$$

We intend to minimize t , so we apply the Euler Lagrange Equation to the integrand to obtain:

$$\frac{d}{dy} \left(\frac{x'}{\sqrt{y(1 - (x')^2)}} \right) = 0 \quad (1.11)$$

This implies that the function in the derivative is constant, which assume to be $1/2a$ to get:

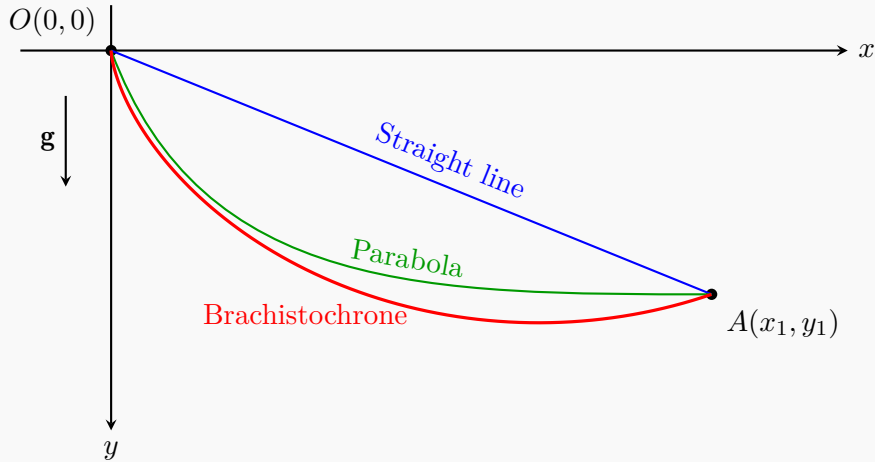
$$x(y) = \int \sqrt{\frac{y}{2a - y}} dy \quad (1.12)$$

This is solved by a substitution of a parameter $y = a(1 - \cos \theta)$ to get:

$$x(\theta) = \int a \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \sin \theta d\theta = a \int 1 - \cos \theta d\theta$$

$$\therefore x(\theta) = a(\theta - \sin \theta) \quad (1.13)$$

The parametric equations $x = a(\theta - \sin \theta)$ and $y = a(1 - \cos \theta)$ represent a *cycloid*, which the path traced by a point on the circumference of a rolling wheel. Another fun fact about this curve is that no matter where you start the ball from, it will always take the same time to reach to the end. In that sense, it is an *isochrone* as well.



The Lagrangian

To apply the Euler Lagrange equation to Classical Mechanics, we start by defining a new quantity called the *Lagrangian* $\mathcal{L}$, which is the difference of the kinetic energy T and potential energy V . It is a function of position ($x_i = x, y, z$), velocity ($\dot{x}_i = \dot{x}, \dot{y}, \dot{z}$) and time (t).

$$\mathcal{L}(x_i, \dot{x}_i, t) = T - V \quad (1.14)$$

For $T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$ and $V = V(x, y, z)$, we consider the following derivatives:

$$\frac{\partial \mathcal{L}}{\partial \dot{x}_i} = m\dot{x}_i \implies \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}_i} \right) = m\ddot{x}_i$$

$$\frac{\partial \mathcal{L}}{\partial x_i} = -\frac{\partial V}{\partial x_i}$$

Using Newton's Law $\mathbf{F} = m\mathbf{a}$, and the definition that $a_i = \ddot{x}_i$ and $F_i = -\partial V / \partial x_i$, we can show that:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}_i} \right) = \frac{\partial \mathcal{L}}{\partial x_i} \quad (1.15)$$

This has the same format as the Euler Lagrange Equation, and this implies the existence of some quantity that is stationary for a Lagrangian that obeys Newton's Law. We call this the *action* S , which is given by:

$$S = \int_{t_1}^{t_2} \mathcal{L}(x_i, \dot{x}_i, t) dt \quad (1.16)$$

Theorem 1.2 – Principle of Stationary Action

The actual path which a particle follows between two points 1 and 2 in a given time interval t_1 to t_2 , is such that the action integral

$$S = \int_{t_1}^{t_2} \mathcal{L} dt \quad (1.17)$$

is stationary when taken along the actual path i.e. $\delta S = 0$. This was given Rowan William Hamilton, and therefore it is sometimes called Hamilton's Principle.

Although we have so far proved this principle only for a single particle and in Cartesian coordinates, we are going to find that it is valid for a huge class of mechanical systems and for almost any choice of coordinates.

Generalized Coordinates

We consider a generalization to N particles, with each having three degrees of freedom which implies $3N$ coordinates to track. We label each coordinate in the cartesian system to be $x_i = x_1, y_1, z_1, \dots, x_N, y_N, z_N$. We define a coordinate system $(q_1, q_2, \dots, q_{3N})$, with q_k as the *generalized position*, and therefore we can write x_i as:

$$x_i = x_i(q_1, q_2, \dots, q_{3N}, t) \quad (1.18)$$

For this to be a valid coordinate system, the Jacobian determinant can't be zero i.e. $\det(\partial x_i / \partial q_k) \neq 0$. The velocity of each coordinate is:

$$\dot{x}_i = \frac{\partial x_i}{\partial q_k} \dot{q}_k + \frac{\partial x_i}{\partial t} \implies \frac{\partial \dot{x}_i}{\partial \dot{q}_k} = \frac{\partial x_i}{\partial q_k} \quad (1.19)$$

We wish to find the Euler Lagrange Equation for this new coordinate system. We start by considering a Lagrangian $\mathcal{L}(q_k, \dot{q}_k, t)$, and we see that:

$$\frac{\partial \mathcal{L}}{\partial q_k} = \frac{\partial \mathcal{L}}{\partial x_i} \frac{\partial x_i}{\partial q_k} + \frac{\partial \mathcal{L}}{\partial \dot{x}_i} \frac{\partial \dot{x}_i}{\partial q_k} \quad (1.20)$$

$$\frac{\partial \mathcal{L}}{\partial \dot{q}_k} = \frac{\partial \mathcal{L}}{\partial \dot{x}_i} \frac{\partial \dot{x}_i}{\partial \dot{q}_k} = \frac{\partial \mathcal{L}}{\partial \dot{x}_i} \frac{\partial x_i}{\partial q_k} \quad (1.21)$$

We "cancel the dots" using the relation that we obtained earlier, and further:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_k} \right) = \underbrace{\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}_i} \right)}_{\partial \mathcal{L} / \partial x_i \text{ by EL eq.}} \frac{\partial x_i}{\partial q_k} + \frac{\partial \mathcal{L}}{\partial \dot{x}_i} \frac{\partial \dot{x}_i}{\partial q_k} \quad (1.22)$$

$$\boxed{\cdot \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_k} \right) = \frac{\partial \mathcal{L}}{\partial q_k}} \quad (1.23)$$

Therefore, we find that the Euler Lagrange Equation for generalized coordinates to be identical to the cartesian one.

Let's introduce further generalizations of other cartesian quantities. We define the *generalized momentum* conjugate to q_k to be:

$$p_k = \frac{\partial \mathcal{L}}{\partial \dot{q}_k} \quad (1.24)$$

And we also define the *generalized force* conjugate to q_k as:

$$\mathcal{F}_k = \frac{\partial \mathcal{L}}{\partial q_k} \quad (1.25)$$

And these two definitions allow us to write the Euler Lagrange equation as:

$$\mathcal{F}_k = \dot{p}_k \quad (1.26)$$

which is identical to Newton's equation $\mathbf{F} = \dot{\mathbf{p}}$.

1.2 Constraints

Types of Constraints

The degrees of freedom of a body are the number of linearly independent ways a particle can move. Point particle in free space have only translational degrees of freedom ($f = 3$), whereas a rigid body can have rotational degrees of freedom as well ($f = 6$). Constraints aim to reduce the degrees of freedom of a body. If there is system of N particles with ℓ constraints, then

the degrees of freedom is $f = 3N - \ell$. This implies that we can get f different generalized coordinates. Constraints can be classified based on their time dependence and existence of constraint equations.

Definition 1.2 – Holonomicity and Time Dependency of Constraints

If there are k constraints which can be written in the following form:

$$f_1(q_1, q_2, \dots, q_f, t) = 0 \quad (1.27)$$

$$f_2(q_1, q_2, \dots, q_f, t) = 0 \quad (1.28)$$

$$\vdots \quad (1.29)$$

$$f_\ell(q_1, q_2, \dots, q_f, t) = 0 \quad (1.30)$$

Then the constraints are called *holonomic*. Example of which are particles confined to move in a circle around some point or If the constraints are in the form of inequalities or any other form, they are said to be *non-holonomic*, with particles confined to move inside a sphere being an example.

The constraints which are time independent are called *scleronomic* constraints are those which are time dependent are called *rheonomic* constraints. Both scleronomic and rheonomic constraints are required to also be holonomic.